Kinvecta

A KINVECTA SERIES

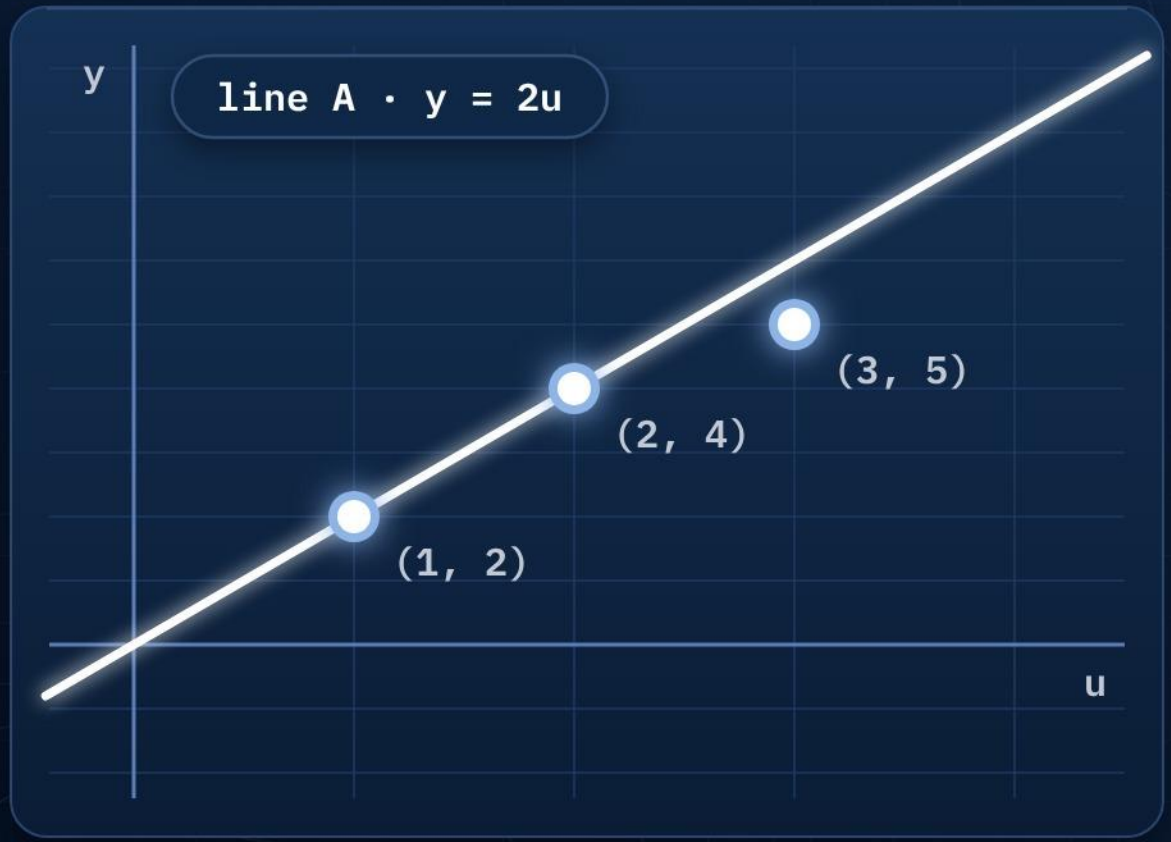
How Large Language Models Work

From math to transformers

EPISODE 10 · GRADIENT DESCENT, PART 2

COLD OPEN · THE EXPERIMENT

NOW CHANGE ONE WORD



LOSS = **SUM** of the 3 penalties

YOUR GUESS

With SUM, the line gets…

better | same | worse

2

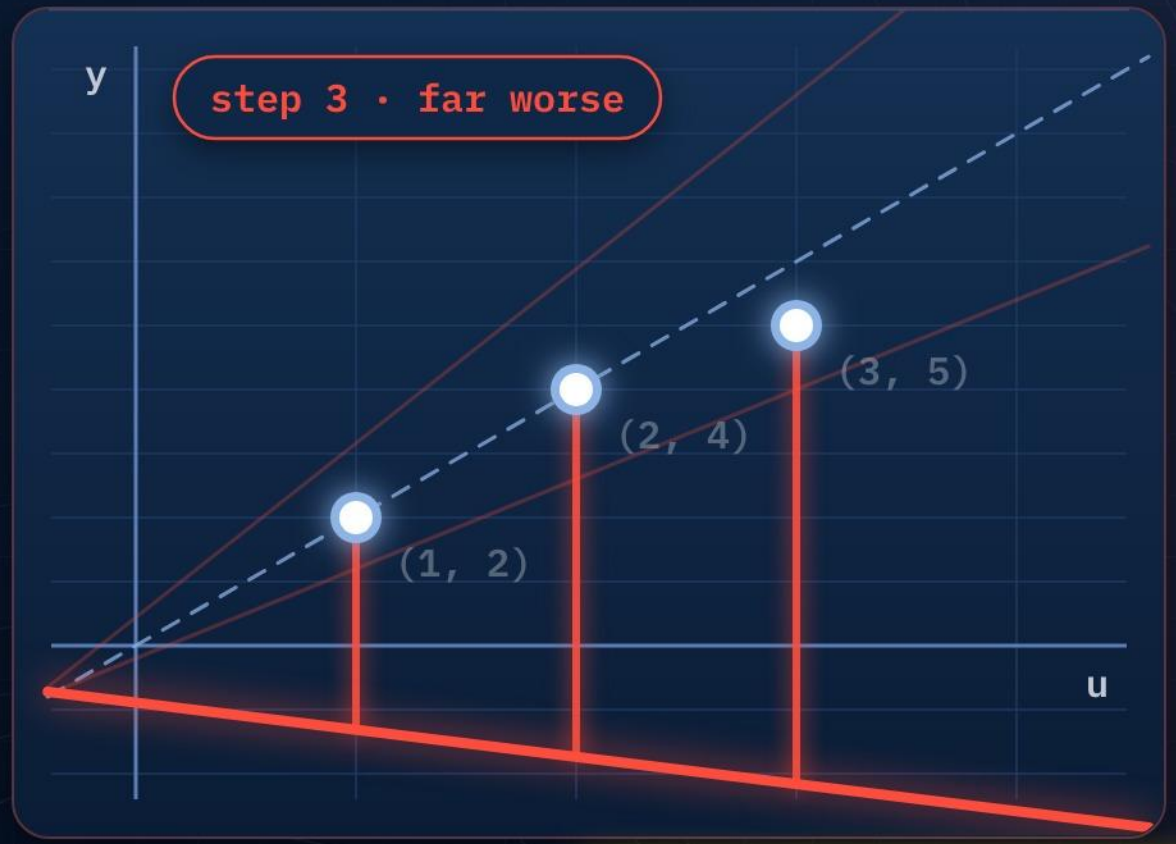
- same dots
- same line
- same step size · 0.1

MARK

COLD OPEN · THE ANSWER

IT GETS WORSE

in good company better same worse



LOSS = SUM of the 3 penalties



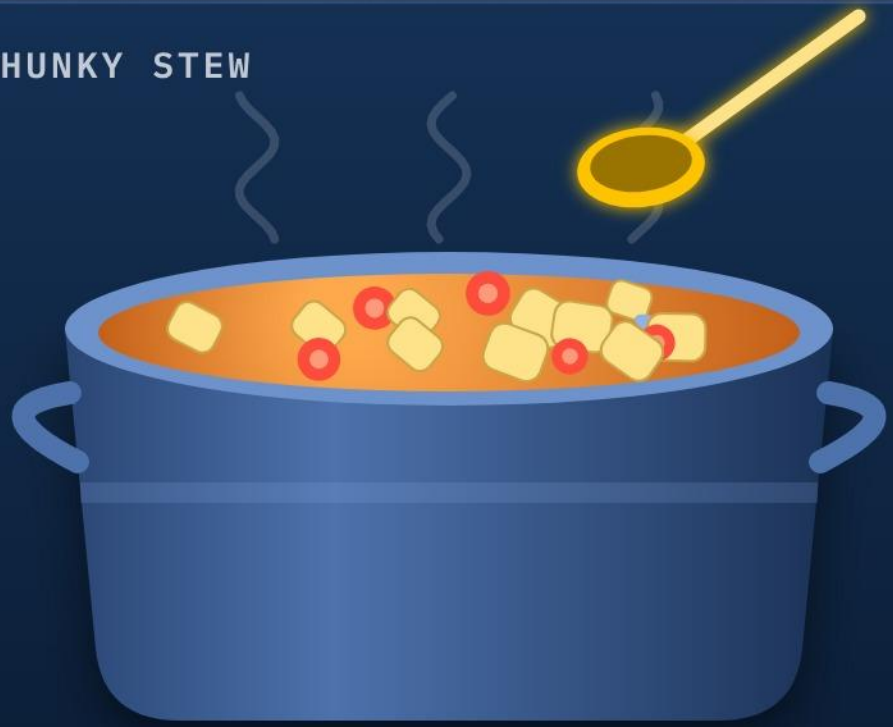
by the end: you'll predict this

MARK

COLD OPEN · PUZZLE 2

LIKE A CHEF TASTING A STEW

A CHUNKY STEW



she doesn't eat the whole pot

SHE TAKES A SPOONFUL

spoonful 1



mostly broth

spoonful 2



mostly potato



why it works

what it costs

taste after taste, she learns the pot

MARK

EPISODE 10 · GRADIENT DESCENT, PART 2

TWO PARTS TODAY

PART 1

The explosion

mean vs **sum**

which step sizes are safe



PART 2

The spoonful



mini-batches

a small handful is enough
worked out by hand





you'll try things first
keep a pencil nearby



every number is in the notebook
linked in the description

MARK

EPISODE 9 · REMINDER

ONE REMINDER WE'LL NEED



z = distance from the bottom

κ = how steep the walls are

η = step size : the learning rate

each step multiplies your distance by

$z \times (1 - 2\eta\kappa)$

smaller than 1 **shrink**

bigger than 1 **fly out**

so $\eta < 1/\kappa$ step size below 1 over kappa

the steeper the valley, the smaller the safe step

MARK

Kinvecta

PART 1 · MEAN VS SUM

MEAN = AVERAGE

THE FACT THAT MAKES TODAY EASY

DOT 1 error^2 penalty + DOT 2 error^2 penalty + DOT 3 error^2 penalty → LOSS = average of the N penalties

W nudge W a little

its own slope + its own slope + its own slope → slope of the loss = average of the slopes

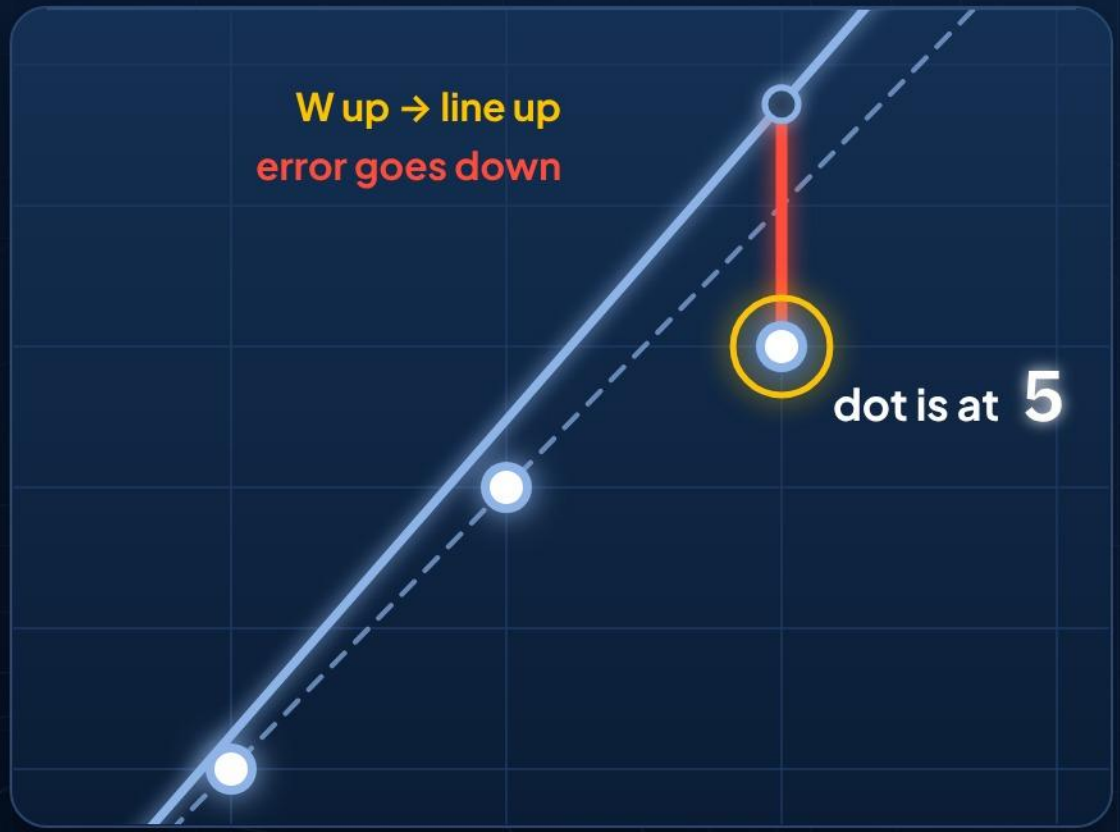
the slope of an average = the average of the slopes

so: find the gradient one dot at a time, then combine

MARK

PART 1 · ONE DOT

THE SLOPE FOR ONE DOT



FIRST: THE ERROR

error = dot's height − line's height

$5 - 6 = -1$ negative = line too high

THE SLOPE FOR W

nudge W: how fast does this dot's penalty change?

$$-2 \times \text{input} \times \text{error}$$

2 from the square · like Episode 9's valley

input W multiplies the input

− raising W lowers the error

PART 1 · THE THIRD DOT

DOT 3: USE THE FORMULA

input = 3 error = −1

slope for W = −2 × input × error

$$-2 \times 3 \times (-1) = 6$$

slope for β = −2 × ~~input ×~~ error

$$-2 \times (-1) = 2$$



dot 3's gradient
= that pair of slopes, one for each knob

W **6** for W

β **2** for β

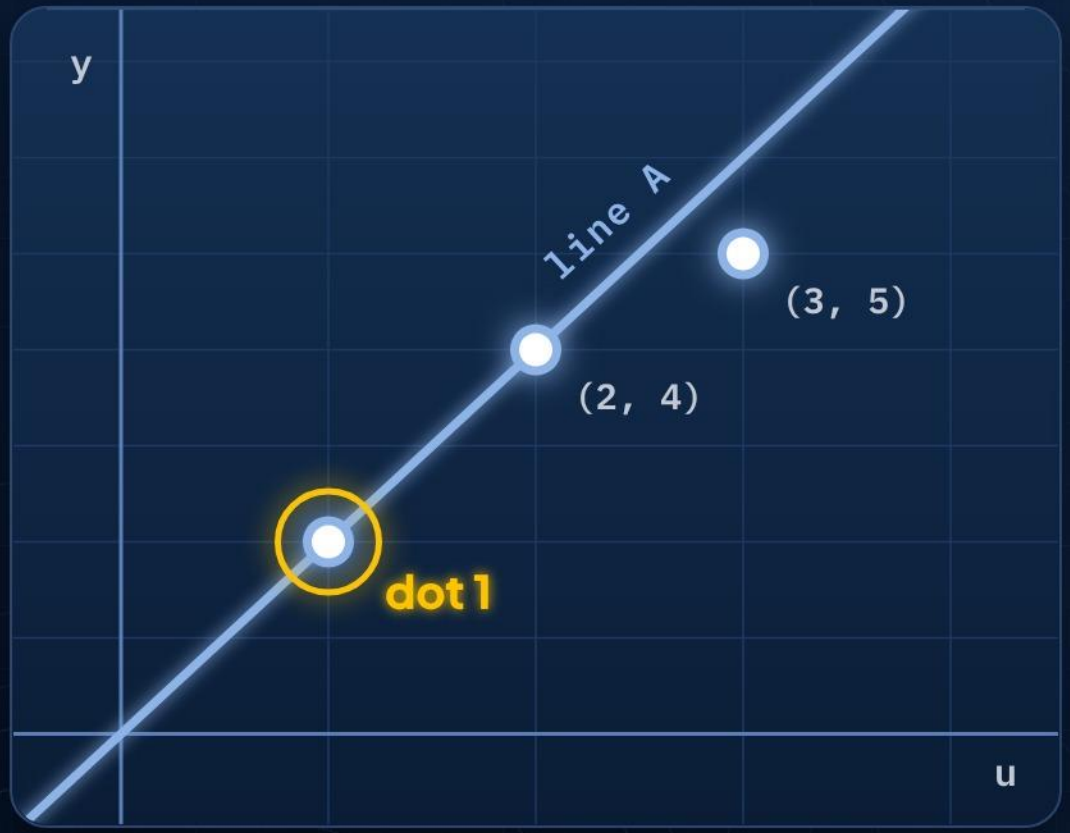
SANITY CHECK

+6 raising W makes it worse

line A is already too high at dot 3

PART 1 · YOUR TURN

NOW YOU: DOT 1



YOUR TURN

Line A passes exactly through dot 1.

its error	slope for W	slope for β
?	?	?

REMINDER

slope for W = −2 × input × error

slope for β = −2 × error

1

PART 1 · ANSWER

ONE GRADIENT PER DOT

DOT 1 · ITS ERROR? ITS SLOPES?

error = 0 the line goes right through it

w −2 × 1 × 0 = 0

β −2 × 0 = 0

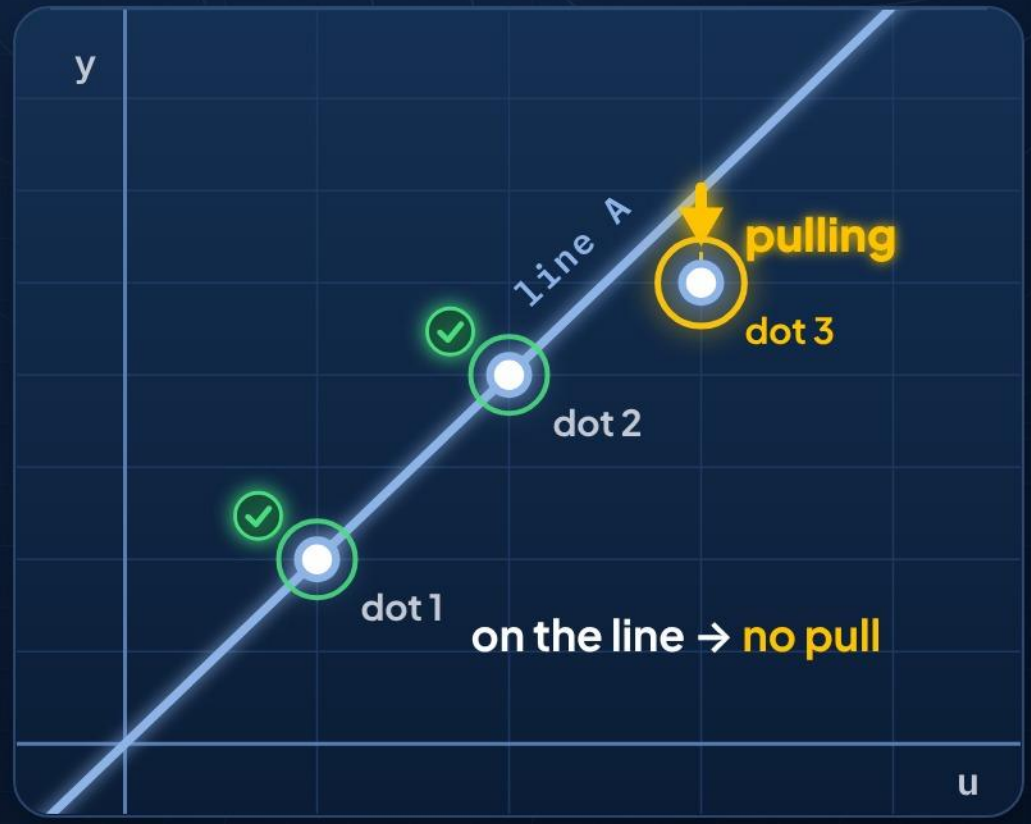
anything × 0 = 0

dot 2: the same story

DOT 1	
w	0
β	0

DOT 2	
w	0
β	0

DOT 3	
w	6
β	2



two dots are happy · only the third is pulling

PART 1 · COMBINE

COMBINE THEM, BOTH WAYS

PER DOT

	W	β
dot 1	0	0
dot 2	0	0
dot 3	6	2

MEAN average the slopes

W $(0 + 0 + 6) \div 3 = 2$

β $(0 + 0 + 2) \div 3 = 2/3$

GRADIENT OF THE MEAN

2 for W, 2/3 for β

SUM just add them

W $0 + 0 + 6 = 6$

β $0 + 0 + 2 = 2$

GRADIENT OF THE SUM

6 for W, 2 for β



same direction

exactly 3× longer

3, because there are 3 dots

PART 1 · THE MEAN STEP

ONE STEP WITH THE MEAN



ONE STEP

η = 0.1

$W = 2 - 0.1 \times 2 = 1.8$

$\beta = 0 - 0.1 \times 2/3 \approx -0.07$

LOSS mean squared error

$0.27^2 + 0.47^2 + (-0.33)^2 \div 3$

0.133 ▼ down from 0.333



a calm step

MARK

YOUR TURN · THE SUM STEP

THE SUM STEP IS YOURS

START · LINE A

$W = 2 \quad \beta = 0$

GRADIENT · THE SUM

6 for W

2 for β

STEP SIZE · STILL

$\eta = 0.1$

YOUR TURN

The same step, with the sum

$\text{new } W = ?$

$\text{new } \beta = ?$

$\text{new} = \text{old} - \eta \times \text{gradient}$

2

work it out

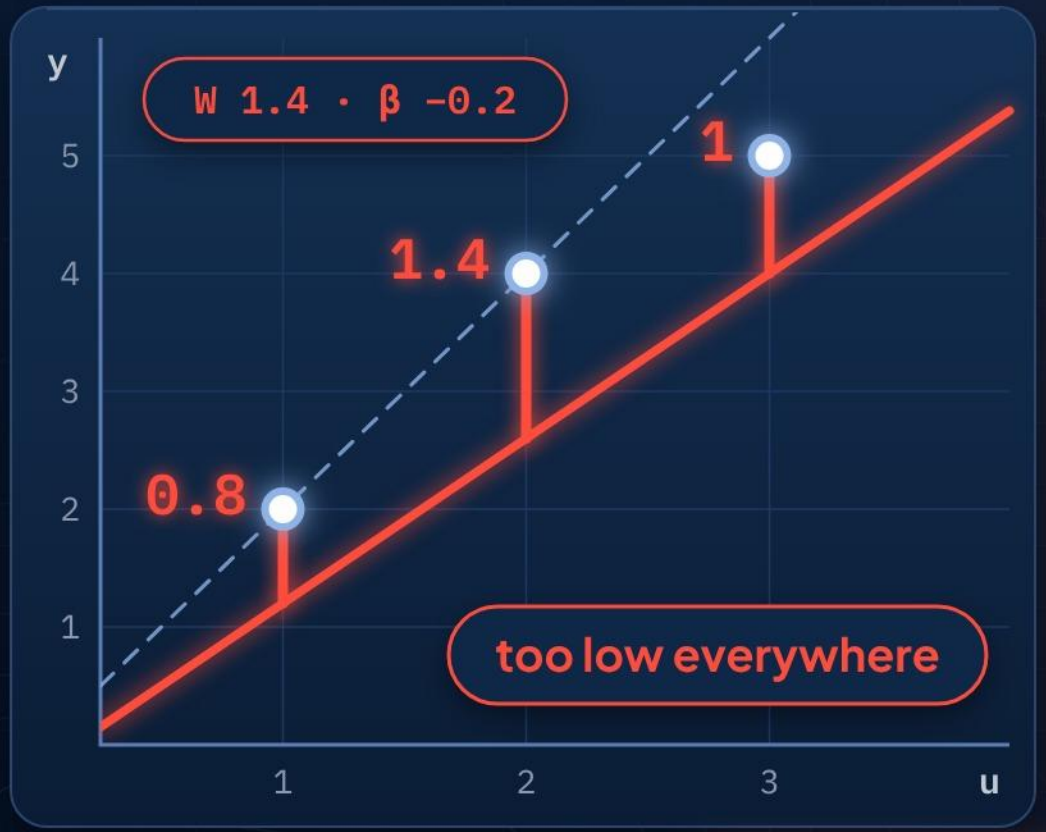
Will this line be…

BETTER or WORSE

MARK

YOUR TURN · THE ANSWER

THE SUM STEP, WORKED OUT



LOSS average of the squares

0.333 → 1.2 ▲ up

THE VALLEY



we flew right over it

PART 1 · THE REASON

WHY DID IT EXPLODE?

EPISODE 9

the steeper the valley,
the smaller the safe step





safe for one

explosive for the other

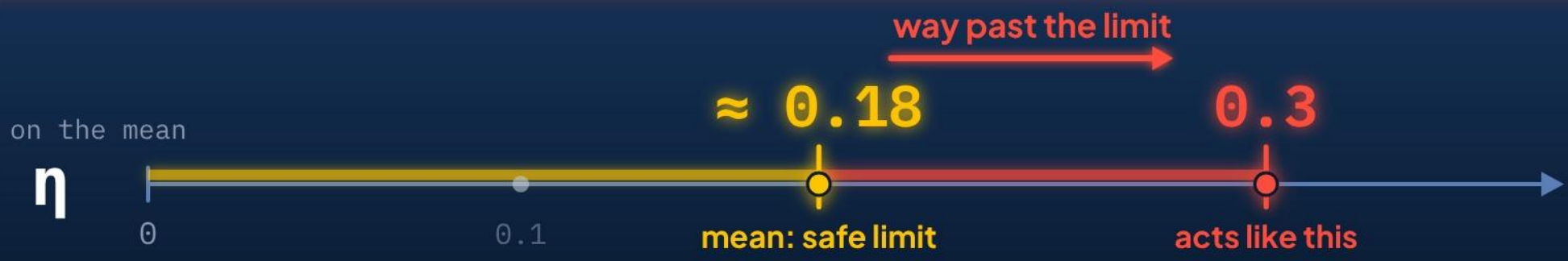
MARK

PART 1 · THE RULE

THE SAME THING, IN ONE LINE

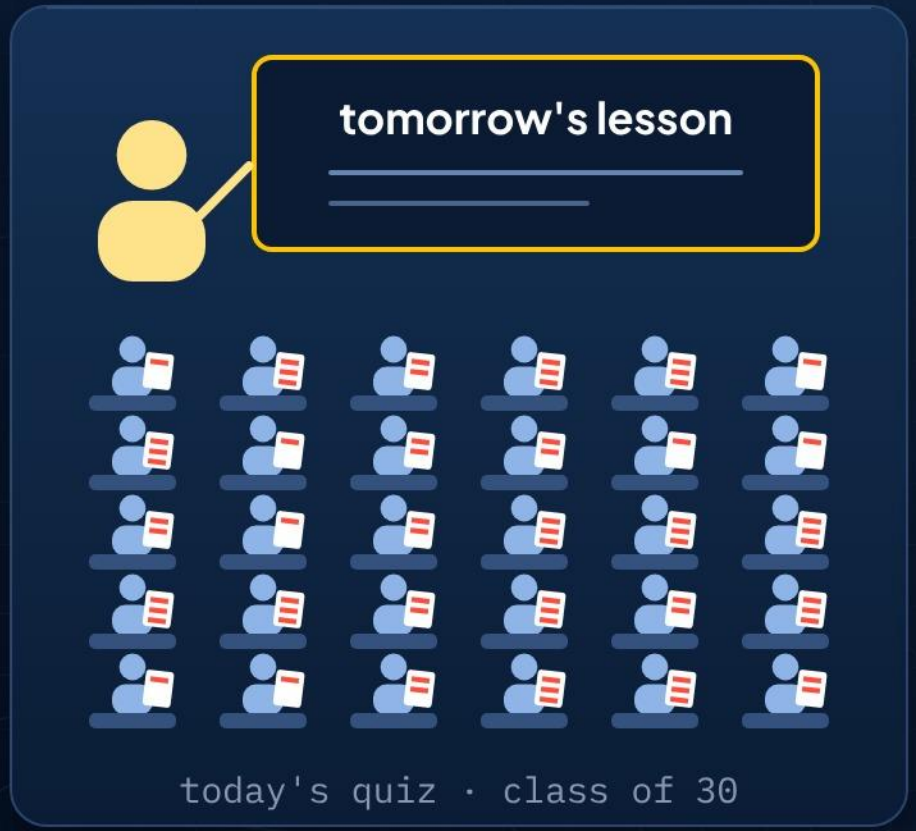
$$\eta \text{ on the SUM} \quad \overset{\text{acts like}}{=} \quad N \times \eta \text{ on the MEAN}$$

0.1 on the sum of 3 dots = **0.3** on the mean



ANALOGY · THE TEACHER

A PICTURE TO REMEMBER IT BY



SHE LOOKS AT THE **CLASS AVERAGE**

sees — a rough day — no re-tuning ✓

corrects → a small correction — the usual default

SHE LOOKS AT THE **CLASS TOTAL** — 30× worse

sees

corrects → ÷ 30

react 30× more gently

the mean: the same step size works when the amount of data changes

MARK

PART 1 · OUT IN THE WILD

MEAN or SUM in real code

```
loss = CrossEntropyLoss(reduction='mean')
```

PyTorch averages by default

cross-entropy = Episode 8's loss

learning rate

η = 0.1

```
loss = CrossEntropyLoss(reduction='sum')
```

but not every piece of code does

η = 0.1

COMMON MISTAKE

copying a learning rate without checking: mean or sum?

same number · completely different step

CHECK · PART 1

QUICK CHECK · four copies of one valley

EPISODE 9'S VALLEY

$f(z) = 3z^2$



$\eta = 0.1$ settled

NOW THE LOSS IS

the SUM of four copies



YOUR TURN

1. the new valley?
2. the new safe limit?
3. what happens at 0.1?

pause here
and try them



Kinvecta

CHECK · PART 1 · ANSWER

HERE'S THE ANSWER

$f(z) = 3z^2$ · loss = the sum of four copies · $\eta = 0.1$

1 $4 \times 3z^2 = 12z^2$

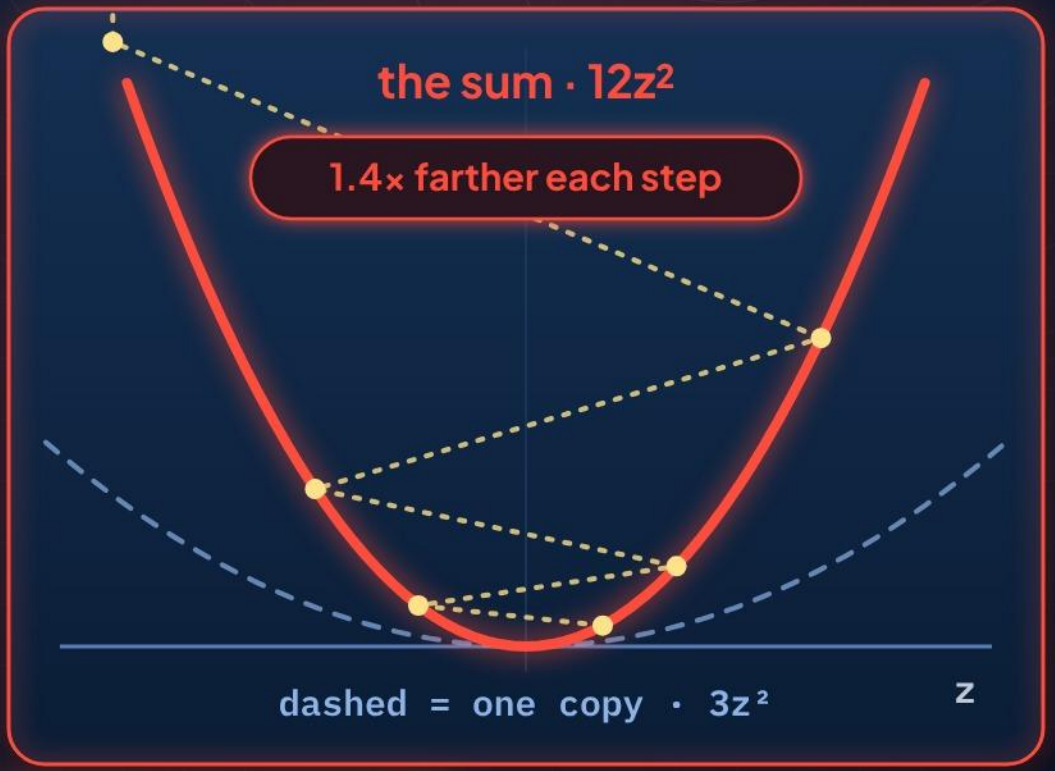
2 $\kappa = 12 \rightarrow 1/12 \approx 0.083$

steepness · safe limit

3 at $\eta = 0.1$: bigger than 1 in size

$2 \times 0.1 \times 12 = 2.4$

$1 - 2.4 = -1.4$



settled on one copy · explodes on the sum of four

MARK

PART 2 · THE PROBLEM

HERE'S THE PROBLEM

A LANGUAGE MODEL · TERMS IN THE LOSS

N = billions



ONE HONEST STEP

= a slope for every single term

step 1

a billion terms for one step…

step 2

…then a billion more for the next

> “Batch gradient descent can be very slow and is intractable for datasets that do not fit in memory”

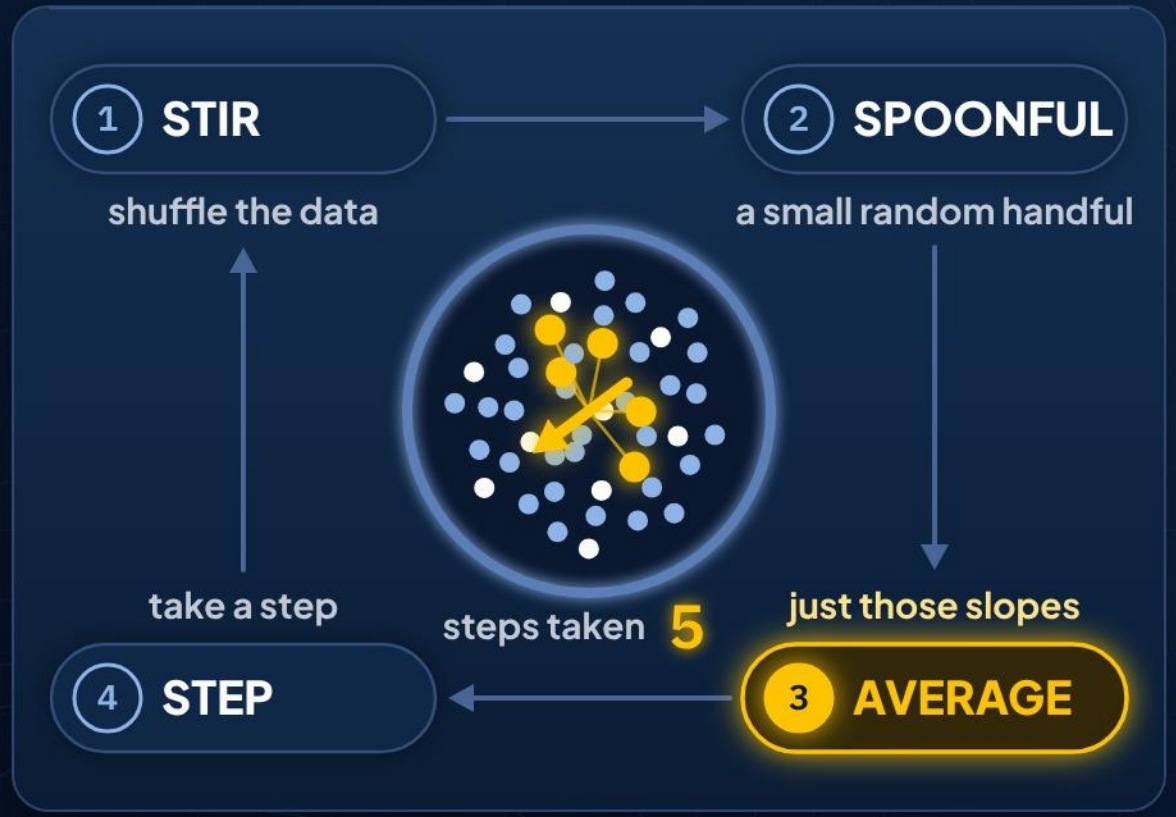
so what do we do instead?

Ruder (2016), §2.1

MARK

PART 2 · WHAT THE CHEF DOES

STIR, SPOONFUL, STEP



MINI-BATCH
= the handful (a small random one)

FULL BATCH
= all N terms together

STOCHASTIC
= random · each step: a random mini-batch

> "Mini-batch gradient descent is typically the algorithm of choice"

Ruder (2016), §2.3

MARK

PART 2 · BUT HOLD ON

A FEW EXAMPLES: wrong direction?

exact direction: found
then one slow, perfect step

	careful	stumbling
a step costs	billions	a handful
steps taken		1,000s of steps



many quick steps beat a few perfect ones

picture after 3Blue1Brown

MARK

CHECK IT BY HAND

LEANING DOWNHILL, BY HAND



dot 1	dot 2	dot 3
0	0	6
W slope	W slope	W slope

full gradient
= their average

= 2

MINI-BATCH OF ONE

one dot, picked at random

draw 1	draw 2	draw 3	never
0	0	6	~~2~~

YOUR TURN

Average many, many draws:
what number do you land on?

1

AVERAGE OF MANY, MANY DRAWS?

THE LONG-RUN AVERAGE IS 2

dot 1	dot 2	dot 3
0	0	6
W slope	W slope	W slope

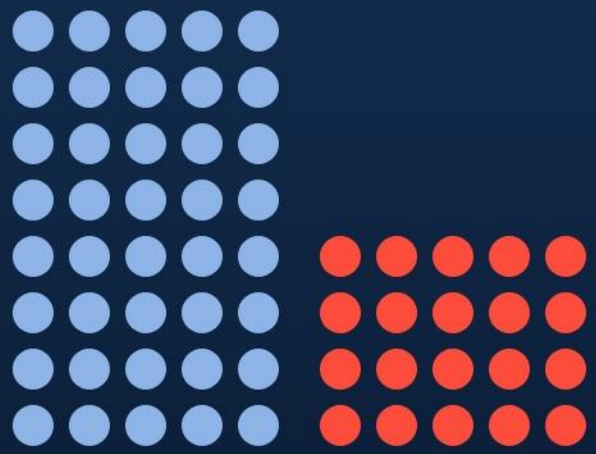
(0 + 0 + 6) / 3 = 2

RUNNING AVERAGE

2.00

after 60 draws

WHAT THE SPOON DREW



slope 0 40

slope 6 20

RUNNING AVERAGE, DRAW BY DRAW



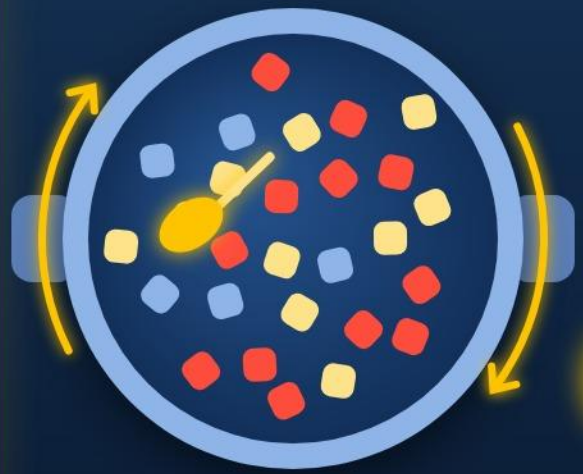
the running average settles right on 2

WHY IT WORKS

THAT'S NOT LUCK

UNBIASED needs RANDOM PICKS

STIRRED POT



spoonfuls



unbiased

SAME CORNER, EVERY TIME



spoonfuls



biased

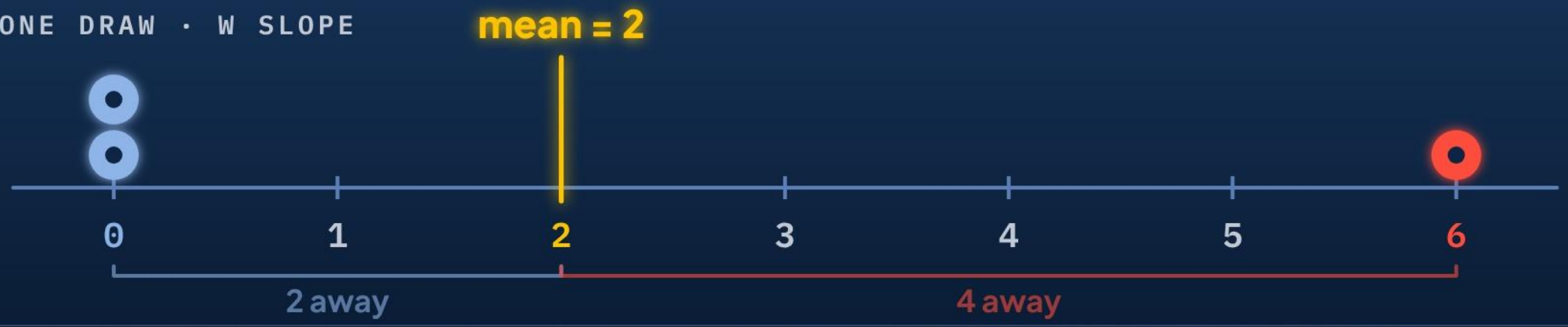
the taste is off

the chef has to stir

THE COST

THE COST: NOISE

ONE DRAW · W SLOPE



VARIANCE = average squared distance from the mean

- 0 — 2 away → $2^2 = 4$
- 0 — 2 away → $2^2 = 4$
- 6 — 4 away → $4^2 = 16$

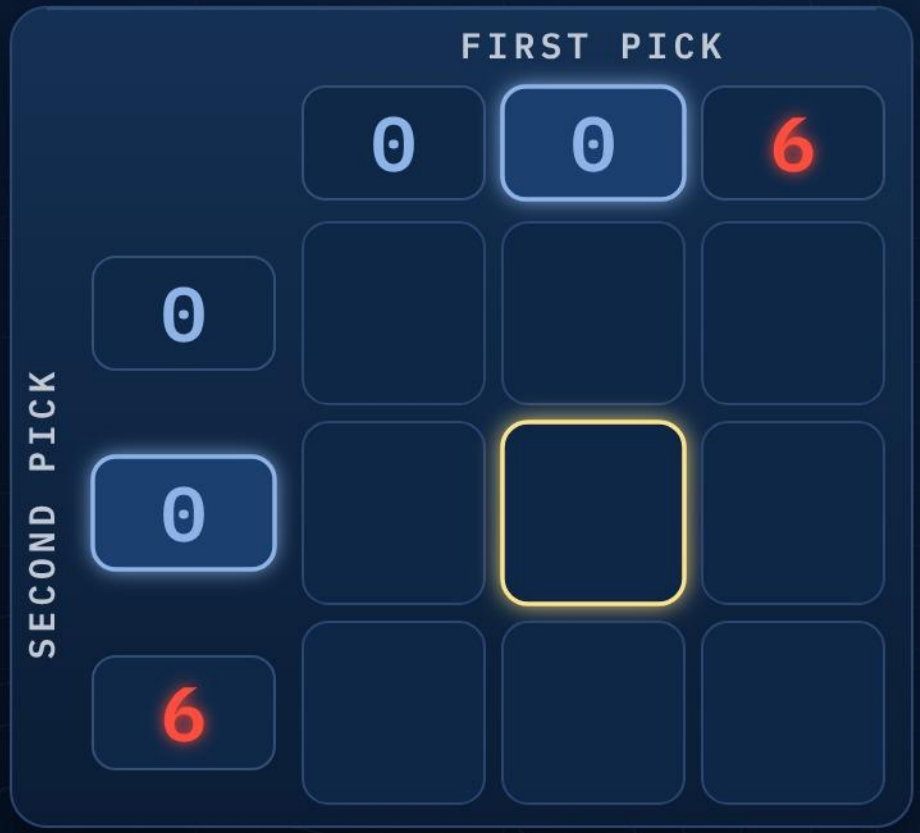
add the squares: $4 + 4 + 16 = 24$

average them: $24 \div 3 = 8$

MARK

MINI-BATCH OF TWO

TWO DOTS PER MINI-BATCH



ONE RULE

put the dot back

the same dot can come up twice

3 choices × 3 choices = 9 pairs

all nine equally likely

YOUR TURN

What averages can a pair give?

How many pairs give each one?

2

MARK

WHAT AVERAGES? HOW MANY PAIRS EACH?

NINE PAIRS, AVERAGED

SECOND PICK \ FIRST PICK	0	0	6
0	0	0	3
0	0	0	3
6	3	3	6

NOW THE VARIANCE

mean = 2

VARIANCE = average squared distance from the mean

- 0 — 2 away → $2^2 = 4$ × 4 = 16
- 3 — 1 away → $1^2 = 1$ × 4 = 4
- 6 — 4 away → $4^2 = 16$ × 1 = 16

16 + 4 + 16 = 36 ÷ 9 = **4**

one dot — variance 8

two dots — variance 4

twice the dots · half the variance

PART 2 · HOW FAST THE NOISE SHRINKS

MORE DOTS, LESS NOISE. HOW MUCH LESS?

VARIANCE

squared units

falls like 1 / batch size



TYPICAL MISS

= √variance



8× the data → the miss didn't even drop to a third

MARK

PART 2 · WHY NOT A BIGGER BATCH?

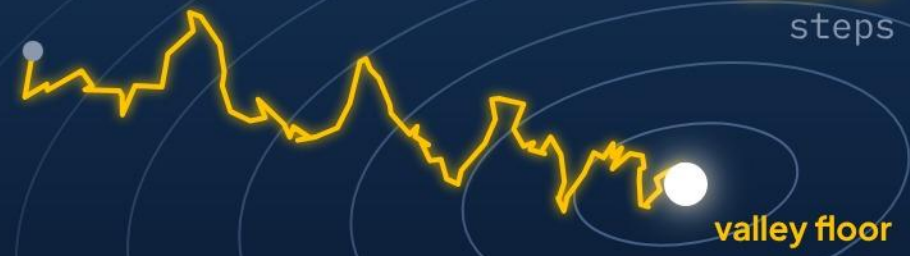
THE SAME POINT, BIGGER NUMBERS

1 PRECISE STEP



100 QUICK STEPS

100 steps



100× the work



for

10× the precision

that's why a spoonful wins

Goodfellow, Bengio & Courville, Deep Learning (2016), §8.1.3

MARK

CHECK · PART 2

YOUR TURN AGAIN

THREE EXAMPLES

their slopes

0	3	9
slope	slope	slope

PAUSE HERE

2

YOUR TURN · WORK IT OUT

1. What's the full gradient (their average)?
2. What's the variance of a mini-batch of one?
3. With two per batch, is the step twice as accurate?

CHECK · PART 2 · THE ANSWER

HERE'S THE ANSWER

slopes 0 · 3 · 9

1 FULL GRADIENT

0 + 3 + 9 = 12

12 ÷ 3 = 4

2 VARIANCE OF ONE DRAW

distances from 4



4^2 1^2 5^2

16 + 1 + 25 = 42

42 ÷ 3 = 14

3 TWO PER BATCH: TWICE AS ACCURATE?

variance 14 → 7

NO

typical miss ≈3.7 → ≈2.6

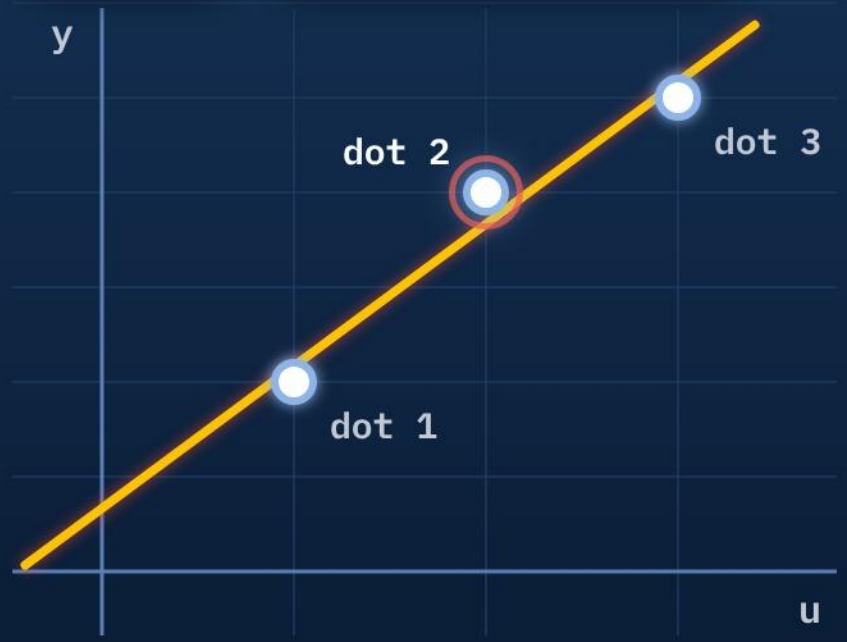
you've understood the real cost of noise

MARK

PART 2 · THE BEST LINE, LINE B

ONE MORE SURPRISE

line B | full gradient = 0



> the noise…
> **does not vanish even when we arrive at a minimum”**
>
> Goodfellow et al. · Deep Learning (2016), §8.3.1

AT THE BOTTOM

…but each step still moves



never quite 0

step = η × slope

shrink η



so the learning rate is lowered over time

Goodfellow et al. (2016), §8.3.1

RECAP

PUT IT TOGETHER

1



gradient of an average
= average of the gradients

2



add instead:
every slope × N

3



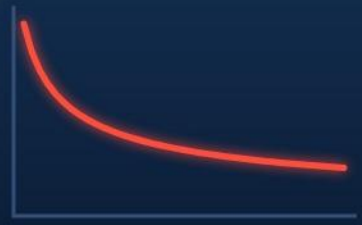
valley N× steeper →
safe step N× smaller

4



a mini-batch:
usually off on one draw,
right on average

5



typical miss shrinks like

$$1 / \sqrt{\text{batch size}}$$

many quick steps beat
a few perfect ones

MARK

RECAP · YOUR TURN

THREE THINGS TO TRY THIS WEEK

01

In the notebook: run line A with the mean and the sum



02

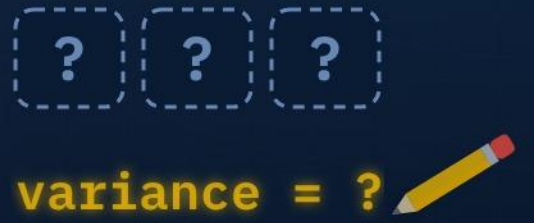
Mini-batches of 1, 2 and 8

average stays at 2 · spread shrinks



03

Make up three slopes work out the variance by hand



THE MISTAKE TO AVOID

Reusing a learning rate without checking whether the loss is a mean

MEAN

companion notebook and every source in the description

NEXT TIME · EPISODE 11

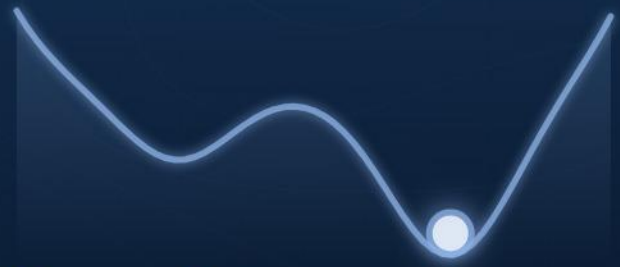
Where gradient descent gets stuck



Saddles
shaped like a potato chip



Valleys
that aren't the deepest



Momentum
a rolling ball gets through

TELL ME IN THE COMMENTS

Did you guess the explosion in the cold open? Be honest.

the notebook and every source are in the description

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How Large Language Models Work

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